

# Federico Melo Barrero

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## EDUCATION

### Universidad de los Andes

Bogotá, Colombia | 2020 – 2024

*B.Sc. Systems and Computing Engineering*

- GPA: 4.92/5.00
- *Summa cum laude*, highest GPA in major in 40 years, highest in Engineering Faculty in 16 years
- Recipient of the Quiero Estudiar scholarship (95% tuition coverage) for the duration of the degree

## AWARDS & HONORS

### Dejar Huella Award

Universidad de los Andes, 2026

*Universidad de los Andes recognition for team impact on the university community. Awarded for "No Estás Solo," a student-built peer-support platform serving 56,000+ active students with ~1,000 daily connections to institutional support*

### Top Saber Pro Score Award 2024

Colombian Ministry of Education, 2025

*Awarded to top-performing students on Colombia's official national standardized higher-education examinations (Saber Pro and TyT)*

### Summa Cum Laude

Universidad de los Andes, 2025

*Summa Cum Laude is the highest undergraduate academic honor. Eligibility requires a cumulative GPA within the top 1% of graduates from the faculty over the past five years, and demonstrated integral merits in an interview before a commission appointed by the Academic Council*

### Athletic Career Distinction

Universidad de los Andes, 2025

*The Athletic Career Distinction is awarded to a single student in the graduating class for exemplary leadership, commitment, and ethics in sports. Awarded citing 41 swimming medals, including 24 gold, and two years as elected team captain*

### Ramón de Zubiría Award (2 times)

Universidad de los Andes; 2023, 2024

*Annual distinction awarded to the undergraduate student with the highest cumulative GPA in their major*

### Semester Excellence Award (3 times)

Universidad de los Andes; Spring 2021, Fall 2021, Fall 2023

*Awarded to the undergraduate student with the highest semester GPA in their academic program during the preceding semester*

### Quiero Estudiar Scholarship

Universidad de los Andes, 2020

*Merit- and need-based scholarship from Universidad de los Andes covering 95% of tuition for the full undergraduate degree*

## WORK EXPERIENCE

### Canals AI

Coral Gables, FL, USA (Remote) | Dec 2024 – Present

*Software Engineer II*

Jul 2025 – Present

- Owned parsing services for the live voice and inquiries agent systems, as part of a pipeline that processed 20000+ orders daily across 115+ wholesale distributors and 9000+ sales reps
- Built the parsing and review layers of a real-time voice service (WebSocket orchestration, speech-to-text, speaker diarization, concurrency control). Extracted ~21 products per call from live phone calls with 75%+ retained through rep review
- Integrated tools (web search, spec sheet retrieval, ERP lookup, order tracking) into an inquiries AI agent that addressed customer questions on orders, materials, and logistics. Achieved 97% of answers citing a source and 80% rated positive by reps
- Contributed to Canals' shared agent framework (Mastra), adding extended thinking, reasoning effort, image/PDF input, temperature, and tool concurrency. Migrated four Parsing agents off hand-rolled, unbounded loops onto bounded runs with step and token ceilings
- Co-built an autonomous agent (Claude Agent SDK) that polled tickets, debugged parsing failures, and shipped prompt-level fixes for review, re-parsing each order to verify its own fix
- One of 4 engineers on the 10-person Parsing team (~60 engineers total) holding merge approval rights on the main branch.

### *Software Engineer I*

Dec 2024 – Jul 2025

- Maintained a parsing pipeline supporting 19 mimetypes with intricate formats (e.g., handwritten, struck-through text, blurry and fragmented photos, cross-references to other files) in a pipeline serving ~60 wholesale distributors processing 4,000+ orders daily
- Abstracted OCR pipeline from single provider to support multiple providers, decoupling the implementation from Textract and enabling format-specific OCR selection, with a cache layer serving ~60% of requests in 0.07s versus 3.6–18.7s uncached
- Led large-scale prompt refactors improving external QA pass rate from 85% to 89%. Involved: model selection, prompt engineering, cache handling, fine-tuning, and regression testing
- Built visual highlighting systems using DOM traversal, coordinate mapping, and string comparison to map parsed products to source documents. Reduced weekly highlighting errors by 2/3
- Joined as the company's 14th engineer.

### **Universidad de los Andes**

Bogotá, Colombia (Part-time) | Jan 2024 – Dec 2024

#### *Full-Stack Developer*

- Designed and built full-stack student data analytics system serving 15,000+ students and 600+ faculty, replacing an external platform and saving ~\$100K USD annually
- Built parallelized data processing pipeline (Python, Polars) with complex filtering and aggregations, exposed through a unified REST API (FastAPI)
- Developed dynamic graphing, advanced filtering, report generation, and role-based dashboards in a React/TypeScript web app
- Built early-warning system for at-risk students (academic performance, dropout indicators), alerting counselors on 1,000+ critical cases
- Architected secure cross-VM communication using air-gapping and network isolation techniques to ensure compliance with student data protection requirements

### **Caseware International Inc.**

Toronto, ON, Canada (Remote, Internship) | Jan 2024 – Jul 2024

#### *Software Developer I*

- Awarded the June Team Award for 'delivering impactful contributions and hard work' as part of the Data Analytics team
- Developed Python-based Jupyter notebooks for financial and auditing data analysis
- Optimized Java and TypeScript microservices performance using AWS S3 and DynamoDB, identifying and resolving bugs
- Authored and delivered technical reports (SPIKEs, RFCs) that influenced team architecture decisions and established best practices

### **Universidad de los Andes**

Bogotá, Colombia (Hybrid, Part-time) | Aug 2023 – Dec 2023

#### *Backend Developer*

- Designed and implemented backend architecture for the Professors' Portfolio system (Java, Spring Boot), a web application for managing faculty academic and administrative information.
- Built persistence, logic, and control layers with CRUD operations, custom logic, and RESTful APIs
- Devised efficient data migration methods to store large volumes of unstructured faculty data into relational databases
- Collaborated with Prof. José Bocanegra, Ph.D. on system architecture documentation and technical specifications, contributing to project planning and implementation roadmaps

## **TEACHING EXPERIENCE**

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### **Universidad de los Andes**

Bogotá, Colombia | Aug 2025 – Present

#### *Adjunct Professor, Algorithm Design*

Fall 2026

- Principal instructor for upper-division algorithm design course, delivering lectures and designing exams
- Topics: divide and conquer, dynamic programming, graph algorithms, NP-completeness, approximation and randomized algorithms

*Adjunct Professor, Introduction to Programming*

Fall 2025, Spring 2026

- Principal instructor for introductory undergraduate course in procedural programming using Python
- Topics: programming fundamentals and data-processing libraries Pandas and Matplotlib

**Universidad de los Andes**

Bogotá, Colombia | Summer 2024

*Undergraduate Teaching Assistant, Vector Calculus*

Supervisor: Prof. Jacinto Puig, Ph.D.

- Designed and graded problem sets, and conducted weekly office hours
- Covered partial derivatives, double and triple integrals, line and surface integrals, vector fields, curl and divergence calculations, and the fundamental theorems (Green's, Stokes', and Gauss')
- Helped students apply these concepts to real-world problems, mainly involving constrained and unconstrained optimization

**Universidad de los Andes**

Bogotá, Colombia | Fall 2022 – Fall 2024

*Undergraduate Research Teaching Assistant, CupiTaller*

Supervisor: Prof. Eduardo Rosales, Ph.D.

- Collaborated in a 7-person team under Prof. Eduardo Rosales, Ph.D., to manage the university's programming support center
- Oversaw the university's Coursera courses 'Programming in Python' and 'Introduction to Object Oriented Programming in Java', which have 50,000+ and 60,000+ historical enrollments respectively
- Managed schedules for ~40 tutors and ~70 TAs each semester, overseeing a system that enabled 1,500+ students to book tutoring sessions
- Designed and co-built CupiHorarios, a scheduling web app (FastAPI, React) leveraging Pyomo (a constraint satisfaction optimization solver) to generate schedules that ensured full coverage during operational hours
- Co-designed and co-built CupiFeedback, a tutoring performance analytics web app (FastAPI, React) that collected feedback from 1,500+ students and presented insights through interactive visualizations
- Co-designed CupiMonitores, a web app that centralized TA evaluation, which was ultimately adopted university-wide
- Developed automation systems including a TypeScript Discord bot for remote tutoring tracking and programming problems with automated solution verification
- Led tutor recruitment and training processes, conducting candidate interviews and occasionally providing coverage during tutor absences, teaching data structures and algorithms concepts

## RELEVANT COURSEWORK

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**Computer Science:** Languages and Automata, Reinforcement Learning, Modeling, Simulation and Optimization, Algorithm Design and Analysis

**Mathematics:** Graph Theory, Game Theory, Computational Methods

**Software Engineering:** Software Metrics and Quality, Software Architecture and Design, API Design and Construction

## PUBLICATIONS

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Martínez Novoa, S., **Melo Barrero, F.**, Ruiz Giraldo, M., Velásquez Marín, S., Carvajal Chaves, N., Hernández Hoyos, M., Reyes, J. P. (2026). "Centralizing the Student Lens: Design and Evaluation of a Learning Analytics Dashboard for Academic Advising in Higher Education". *Proceedings of the World Engineering Education Forum (WEEF 2026)*. IFEEES & GEDC. To appear.

**Melo Barrero, F.** (2024). "Desarrollo del Perfil del estudiante dentro de No estás solo". *Undergraduate thesis*. Universidad de los Andes. Available at: [Uniandes Repository](#)

## TECHNICAL EXPERIENCE

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**AI Engineering & LLM Orchestration:** Anthropic / OpenAI / Gemini / Groq APIs, Claude Agent SDK, Mastra, MCP

**Backend & Systems Architecture:** Python, Java, TypeScript, FastAPI, Spring Boot, WebSocket  
Orchestration, Concurrency Control

**Data Engineering & Infrastructure:** Polars, PostgreSQL, Redis, AWS (RDS, ECS, S3, DynamoDB),  
Docker, GitHub CI/CD

**Product & Interface:** React, Next.js, Svelte, Plotly, ShadCN / MUI

## EXTRACURRICULAR ACTIVITIES

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### **National-level swimmer**

- Member and captain of the Universidad de los Andes Swimming Team, earning 24 gold, 10 silver, and 7 bronze medals; four-time member of the Bogotá City Swimming Team, competing in 12 National Swimming Championships
- 1st place, men's 20–29, 2 km open water, OCEANMAN San Andrés 2026

## ADDITIONAL INFORMATION

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**Languages:** Spanish (native), English (IELTS Academic 8.5 / CEFR C2)